

JEE Advanced Booster Test - 2 | 2022

Date: 09/11/2020

Maximum Marks: 180

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

- The question paper consists of 3 Parts (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **four** sections (**Section 1, Section 2, Section 3 and Section 4**).
- Section 1** contains **6 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

Section 2 contains **2 Paragraphs containing 2 Questions** on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

Section 3 contains **2 Column-Match sets**. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

Section 4 contains **6 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION - 1

- This section consists of **SIX (06)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+4 If only (all) the correct option(s) is(are) chosen
Partial Marks:	+3 If all the four options are correct but ONLY three options are chosen
Partial Marks:	+2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks:	+1 If two or more options are correct but ONLY one option is chosen, and it is a correct option
Zero Mark:	0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION - 2

- This section has **TWO (02) Paragraphs containing TWO (02) Questions** on each paragraph. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the correct option is chosen
Zero Marks:	0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION - 3

- This section has **TWO (02) Column-Match sets**. Each set contains statements given in two columns. Statements in column I have to be matched with statements in column II. One or more than one statements in column II can be matched with a statement in column I.
- Each set has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the option corresponding to the correct combination is chosen
Zero Marks:	0 If none of the options is chosen (i.e., the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION 4

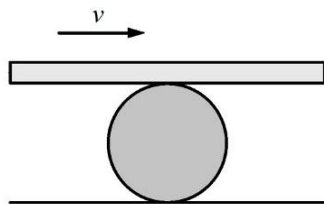
- This section contains **6 Integer Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the correct Integer value is entered. There is NO negative marking .
Zero Marks:	0 In all other cases.

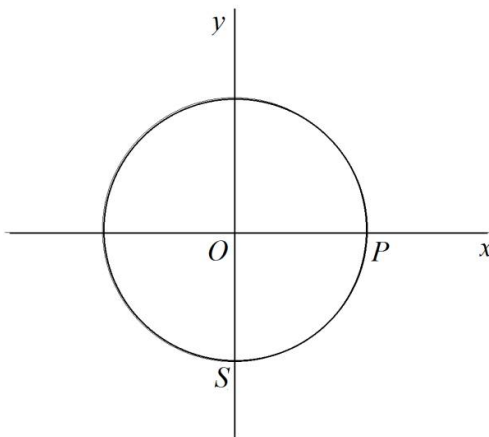
PART I : PHYSICS**60 MARKS****SECTION 1**

This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- Disc A of radius r moving with speed 5 m/s on a smooth horizontal surface hits another disc B of radius $2r$ lying at rest. At the moment of impact, the velocity vector of centre of disc A is at a perpendicular distance $1.8r$ from centre of disc B. It is observed that discs move perpendicular to each other after impact. Mass of A is 1 kg and mass of B is 4 kg , both discs are smooth. Choose the correct options.
 - Velocities of A and B after impact are $V_A = 4 \text{ m/s}$ and $V_B = 0.75 \text{ m/s}$
 - Velocities of A and B after impact are $V_A = 3 \text{ m/s}$ and $V_B = 1 \text{ m/s}$
 - The collision is perfectly elastic
 - The collision is not elastic with coefficient of restitution 0.25
- A solid cylinder of radius R and mass M is rolled on a horizontal surface with a plank such that the plank remains horizontal and moves with a horizontal velocity v as shown. There is no slipping between the plank and the cylinder, and no slipping between the cylinder and the surface. The angular velocity and kinetic energy of the cylinder is ω and K respectively. Which of these options is/are correct?

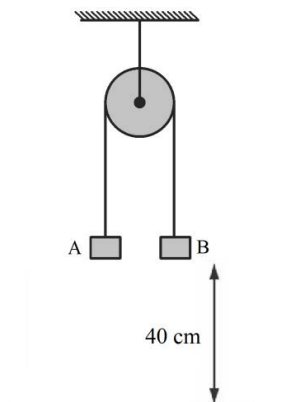


- $\omega = \frac{v}{R}$
 - $\omega = \frac{v}{2R}$
 - $K = \frac{3}{16} M v^2$
 - $K = \frac{5}{16} M v^2$
- A particle is moving in a circular path in the X-Y plane centred at the origin with constant kinetic energy K and time period T . At $t = 0$, it is at the point P on the positive X-axis and moving towards the +Y direction. At $t = \frac{T}{4}$, let its angular momentum about the origin be $\vec{L}_O$ and let its angular momentum about the point S on the negative Y-axis (shown in the figure) be $\vec{L}_S$. Which of these options is/are correct?

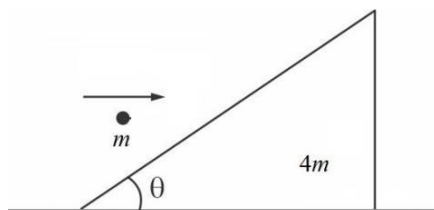


- $\vec{L}_O = \frac{KT}{2\pi} \hat{k}$
- $\vec{L}_O = \frac{KT}{\pi} \hat{k}$
- $\vec{L}_S = \frac{KT}{\pi} \hat{k}$
- $\vec{L}_S = \frac{2KT}{\pi} \hat{k}$

4. Two blocks A and B of mass 1.9 kg and 2.1 kg are suspended using an ideal, massless pulley and string as shown. Initially, both blocks are at a height 40 cm from the ground. The system is released from rest at $t = 0$. At $t = 1.0$ s, the string suddenly breaks at the point it is attached to block B. At $t = 1.1$ s, the velocity of the blocks A and B are v_A and v_B respectively and their height above the ground is h_A and h_B respectively. Which of these options is/are correct? ($g = 10 \text{ m/s}^2$)

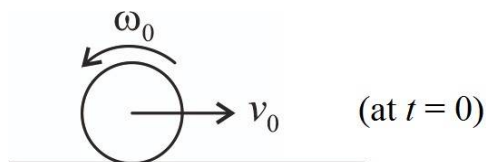


- (A) $v_A = 0.5 \text{ m/s}$, downward (B) $v_B = 1.5 \text{ m/s}$, downward
(C) $h_A = 25 \text{ cm}$ (D) $h_B = 5 \text{ cm}$
5. A particle of mass m collides with a movable inclined plane of mass $4m$, which was initially at rest on a horizontal surface. There is no friction anywhere. Just before the collision, the particle was travelling horizontally with velocity u and just after the collision, it is travelling vertically with velocity v and the inclined plane is travelling with velocity v_p . Which of these options is/are correct?



- (A) $v = u \cot \theta$ (B) $v = u \tan \theta$ (C) $v_p = \frac{u}{4}$ (D) $v_p = \frac{u}{5}$
6. A uniform solid cylinder of mass 0.5 kg and radius 10 cm moves on a uniformly rough horizontal surface with coefficient of friction 0.1. At $t = 0$, the velocity of the centre of mass and angular velocity of the cylinder are respectively $v_0 = 1 \text{ m/s}$ and $\omega_0 = 50 \text{ rad/s}$ in the directions shown. At $t = 3 \text{ s}$, the velocity of the centre of mass and the angular velocity of the cylinder are respectively v and ω . Which of the following options is/are correct?

($g = 10 \text{ m/s}^2$)



- (A) $v = 1 \text{ m/s}$, towards left (B) $v = 2 \text{ m/s}$, towards left
(B) $\omega = 10 \text{ rad/s}$, clockwise (D) $\omega = 10 \text{ rad/s}$, anti-clockwise

SPACE FOR ROUGH WORK

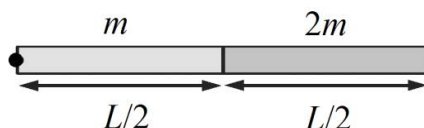
SECTION 2

This section has **TWO (02) Paragraphs** containing **TWO (02) Questions** on each paragraph. Each question has **FOUR options**. **ONLY ONE** of these four options is the correct answer.

Paragraph for Q.7 to Q.8

Two thin rods of length $\frac{L}{2}$ each, and mass m and $2m$ are joined end to end to form a composite rod of length L .

The composite rod is pivoted at the left end so that it can rotate in the vertical plane, and then released from the horizontal position as shown. At the instant the rod becomes vertical, its kinetic energy is K and its angular velocity is ω .



7. K is equal to:
- (A) $\frac{7}{2}mgL$ (B) $\frac{9}{2}mgL$ (C) $\frac{7}{4}mgL$ (D) $\frac{9}{4}mgL$
8. ω is equal to:
- (A) $\sqrt{\frac{18g}{5L}}$ (B) $\sqrt{\frac{9g}{5L}}$ (C) $\sqrt{\frac{14g}{5L}}$ (D) $\sqrt{\frac{7g}{5L}}$

Paragraph for Q.9 to Q.10

A spaceship of mass M moves in a gravity-free region with velocity v . The occupants of the spaceship spot an stationary object (say object X) directly ahead in its path, and so a projectile of mass $\frac{M}{200}$ is fired with velocity $4v$ from the spaceship towards the object X . All velocities are relative to a frame fixed in the free space.

9. After the projectile is fired, the velocity of the spaceship becomes:
- (A) $\left(\frac{196}{200}\right)v$ (B) $\left(\frac{197}{200}\right)v$ (C) $\left(\frac{195}{199}\right)v$ (D) $\left(\frac{196}{199}\right)v$
10. The projectile collides with the object X , gets embedded in it, and the combined system then moves at velocity $\frac{v}{3}$. Then, the mass of the object X is:
- (A) $\left(\frac{11}{200}\right)M$ (B) $\left(\frac{5}{200}\right)M$ (C) $\left(\frac{8}{199}\right)M$ (D) $\left(\frac{14}{199}\right)M$

SPACE FOR ROUGH WORK

SECTION 3

This section contains 2 Column-Match sets. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

11. List-I below contains forces, some of which vary with the position of a particle (x, y) .

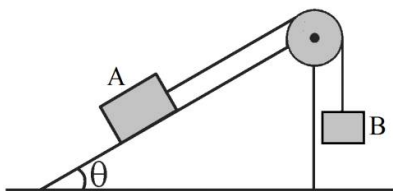
A particle travels along a straight line path from the point $A(-1, -1)$ to the point $B(1, 1)$ under the action of these forces. Match each force with the work done by the force on the particle from List-II.

All quantities (force, distance, work) are in SI units.

LIST-I		LIST-II	
(p)	$\vec{F} = \hat{i} + \hat{j}$	(i)	0
(q)	$\vec{F} = \hat{i} - 2\hat{j}$	(ii)	-2
(r)	$\vec{F} = 3y^2\hat{i}$	(iii)	2
(s)	$\vec{F} = x\hat{i} + y\hat{j}$	(iv)	4

- (A) (p) – (iv), (q) – (i), (r) – (i), (s) – (i) (B) (p) – (iv), (q) – (iii), (r) – (ii), (s) – (i)
 (C) (p) – (i), (q) – (iii), (r) – (i), (s) – (ii) (D) (p) – (iv), (q) – (ii), (r) – (iii), (s) – (i)

12. Two blocks A and B of mass $m_1 = 2 \text{ kg}$ and $m_2 = 1 \text{ kg}$ are arranged with a fixed inclined plane and ideal, massless pulley and string as shown. The angle $\theta = \sin^{-1}(0.6)$. The coefficient of friction between the plane and block A is μ . The system is released from rest. Match each possible value of μ in List-I to the correct outcome from List-II.



LIST-I		LIST-II	
(p)	$\mu = 0.05$	(i)	The blocks are in equilibrium
(q)	$\mu = 0.1$	(ii)	The blocks move with acceleration 0.4 m/s^2 such that block B moves up
(r)	$\mu = 0.2$	(iii)	The blocks move with acceleration 0.53 m/s^2 such that block B moves down
(s)	$\mu = 0.25$	(iv)	The blocks move with acceleration 0.13 m/s^2 such that block B moves up

- (A) (p) – (ii), (q) – (iv), (r) – (iii), (s) – (i) (B) (p) – (ii), (q) – (iv), (r) – (i), (s) – (i)
 (C) (p) – (iii), (q) – (iii), (r) – (ii), (s) – (i) (D) (p) – (iii), (q) – (iv), (r) – (ii), (s) – (ii)

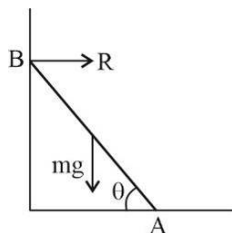
SPACE FOR ROUGH WORK

SECTION 4

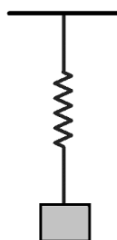
This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.*

(Example: 6, 81, 1.50, 3.25, 0.08)

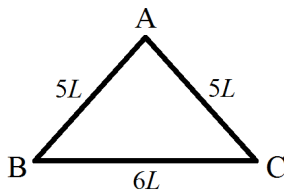
13. A thin rod of mass 2 kg rests in equilibrium against a smooth vertical wall, making an angle $\theta = \tan^{-1}\left(\frac{4}{3}\right)$ with the rough horizontal ground as shown. The force exerted by the rod on the wall is _____ N. ($g = 10 \text{ m/s}^2$)



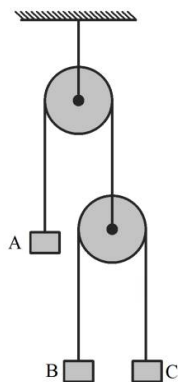
14. A block of mass 1 kg is suspended vertically by an ideal spring of spring constant 100 N/m as shown, and the upper end of the spring is fixed. The block is released from rest when the spring is in its natural length. The velocity of the block after it has moved down by a distance 10 cm is _____ m/s. ($g = 10 \text{ m/s}^2$)



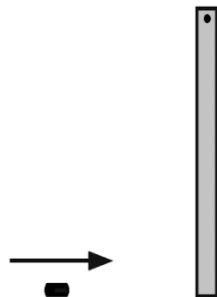
15. A thin uniform wire of length $16L$ is bent to form an isosceles triangle ABC, such that $AB = AC = 5L$ and $BC = 6L$. If the distance of the centre of mass of this triangle from A is βL , the value of β is _____.



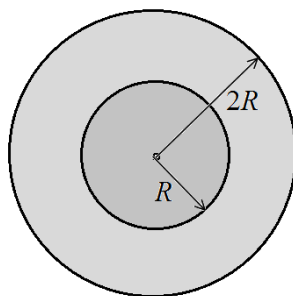
16. Three blocks A, B and C are arranged with ideal, massless pulleys and strings as shown. The mass of blocks B and C is 1 kg and 3 kg respectively. The system is released from rest. For the block A to remain at rest, its mass must be _____ kg. ($g = 10 \text{ m/s}^2$)



17. A thin uniform rod of mass $2m$ and length L , hinged at its upper end, hangs in equilibrium. A small bullet of mass m is fired horizontally with velocity v . The bullet hits the rod very close to its bottom end and gets embedded in it. The angular velocity of the combined system just afterwards is $X \left(\frac{v}{L} \right)$. The value of X is _____.



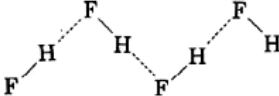
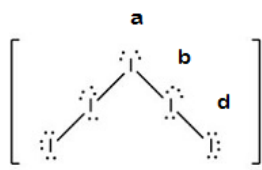
18. From a thin, uniform circular disc of radius $2R$ and mass per unit area σ_0 , a smaller concentric disc of radius R is cut out. Now a disc of radius R and mass per unit area $2\sigma_0$ is fixed inside the space created, resulting in the composite disc shown in the figure. If the composite disc is rotated about a diameter with angular velocity ω , its kinetic energy is $\frac{X}{8} (\pi \sigma_0 R^4 \omega^2)$. The value of X is _____.



SPACE FOR ROUGH WORK

PART II : CHEMISTRY**60 MARKS****SECTION 1**

This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- Which of the following is/are correct for O_3 , I_3^+ , I_3^- and C_3^{4-} ?
 - I_3^- and C_3^{4-} are linear in shape
 - O_3 and I_3^+ are angular in shape
 - $C_3^{4-} < O_3 < I_3^-$ (order of lone pair electron on central atom)
 - $C_3^{4-} < O_3 < I_3^-$ (order of p-character in hybrid orbital of central atom)
- Which of the following is/are correct for diamond and graphite?
 - Diamond < Graphite : order of C – C bond dissociation energy
 - Diamond < Graphite : order of ΔH_f°
 - Diamond < Graphite : order of thermodynamic stability
 - Diamond < Graphite : order of ΔH_{comb}°
- Which of the following is/are correct for HF?
 - HF is diamagnetic : Acc. to molecular orbital theory
 - HF and B_2 have same type of bond : Acc. to molecular orbital theory
 -  : Hydrogen bonding in liquid HF
 - $HF > H_2O$: Order of boiling point
- Which of the following is/are TRUE about critical point (Pc, Tc) in the P-V diagram?
 - Gas, liquid and solid states exist in equilibrium at the critical point
 - Density of a liquid and its vapour become equal at the critical point
 - Interface between a liquid and its vapour disappears at the critical point
 - An ideal gas system cannot exhibit a critical point
- Which of the following is/are correct for structure given below for $[I_5]^-$?
 - There are 14 lone pairs of electrons in given structure for $[I_5]^-$
 - There are 5 σ bonds in given structure for $[I_5]^-$
 - I(a) has positive formal charge while I(b) has negative formal ch
 - I(a) is sp^3 hybridized while I(b) is sp^3d hybridized
- Which of the following is/are true?
 - U is a state function
 - H is a state function
 - $(q + w)$ is a state function
 - w is a path function

SECTION 2

This section has **TWO (02) Paragraphs** containing **TWO (02) Questions** on each paragraph. Each question has **FOUR options**. **ONLY ONE** of these four options is the correct answer.

Paragraph for Q-7 to Q-8

The physical properties melting point, boiling point, vapour pressure, evaporation, viscosity, surface tension, and solubility are related to the strength of attractive forces between molecules. These attractive forces are called **Intermolecular Forces**. The amount of “stick togetherness” is important in the interpretation of the various properties listed above.

7. Which gas has the smallest molar volume at STP?
 (A) He (B) NH₃ (C) N₂ (D) CO
8. Which liquid has the highest vapour pressure at 25°C?
 (A) CHCl₃ (B) CFCl₃ (C) CCl₄ (D) CBrCl₃

Paragraph for Q-9 to Q-10

The **standard enthalpy of formation** or **standard heat of formation** of a compound is the change of enthalpy during the formation of 1 mole of the substance from its constituent elements, with all substances in their standard states.

The standard enthalpy change of any reaction can be calculated from the standard enthalpies of formation of reactants and products.

9. For which reaction is ΔH_{rxn}^0 equal to ΔH_f^0 of CaSO₃(s)?
 (A) $\text{Ca(s)} + \frac{1}{8}\text{S}_8\text{(s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{CaSO}_3\text{(s)}$
 (B) $8\text{Ca(s)} + \text{S}_8\text{(s)} + 12\text{O}_2\text{(g)} \rightarrow 8\text{CaSO}_3\text{(s)}$
 (C) $8\text{Ca(s)} + \text{S}_8\text{(g)} + 12\text{O}_2\text{(g)} \rightarrow 8\text{CaSO}_3\text{(s)}$
 (D) $\text{CaO(s)} + \text{SO}_2\text{(g)} \rightarrow \text{CaSO}_3\text{(s)}$
10. If the average carbon-hydrogen bond dissociation enthalpy in ethane is 416 kJ mol⁻¹, what is the bond dissociation enthalpy of the carbon-carbon bond in ethane?

Species	ΔH_f^0 , kJ mol ⁻¹
C ₂ H ₆ (g)	-84.7
H(g)	217.9
C(g)	718.4

- (A) 164 kJ mol⁻¹ (B) 333 kJ mol⁻¹ (C) 386 kJ mol⁻¹ (D) 404 kJ mol⁻¹

SPACE FOR ROUGH WORK

SECTION 3

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11.

Property		Type/Effect of temperature of pressure	
I.	Surface Tension of water	P	Intensive property
II.	Boiling Point of water	Q	Extensive property
III.	Density of water	R	Decrease with increase in temperature
IV.	Aqueous Tension of water	S	Decrease with decrease in pressure

(A) [I-P, R] [II-P, S] [III-P, R] [IV-P]

(B) [I-Q, R]; [II-Q, S]; [III-Q, R]; [IV-P]

(C) [I-P, S]; [II-Q]; [III-P, R]; [IV-P]

(D) [I-Q, S]; [II-P, S]; [III-P, R]; [IV-P]

12.

	Molecular species		Properties
I	H ₂ O	P	$\Delta H_f^\circ = 0$
II	H ₂ O ₂	Q	sp ³ hybridized central atom
III	HO ₂	R	paramagnetic
IV	O ₂	S	polar
V	O ₃	T	sp ² hybridized central atom

(A) [I-P, S, T]; [II-R, S, T]; [III-P, R, T]; [IV-P, R]; [V-Q, R, S]

(B) [I-Q, S]; [II-Q, S]; [III-Q, R, S]; [IV-P, R]; [V-S, T]

(C) [I-P, Q, S]; [II-Q, R, S]; [III-Q, R, S]; [IV-P, R]; [V-R, S, T]

(D) [I-R, S, T]; [II-P, S, T]; [III-P, R, T]; [IV-P, R]; [V-Q, R, S]

SPACE FOR ROUGH WORK

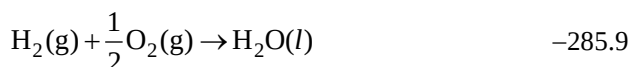
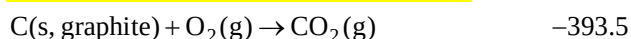
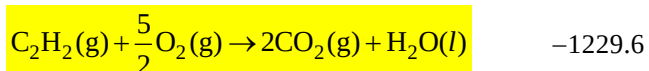
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This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.*

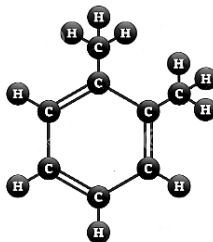
(Example: 6, 81, 1.50, 3.25, 0.08)

13. What is the standard enthalpy of formation of $\text{C}_2\text{H}_2(\text{g})$ in kJ mol^{-1} ?

$$\Delta H_{\text{rxn}}^{\circ}, \text{kJ mol}^{-1}$$



14. A 10.0 g sample of solid NH_4NO_3 ($M = 80.05$) is added to 100.0 g H_2O in a well-insulated container. Both the solid and the water are initially at 24.00°C , but after the NH_4NO_3 dissolves, the temperature of the mixture is 17.11°C . What is $\Delta H_{\text{rxn}}^{\circ}$ (in kJ mol^{-1}) for the dissolution of $\text{NH}_4\text{NO}_3(\text{s})$? Assume that the specific heat capacity of the NH_4NO_3 solution is the same as that of pure water, $4.184 \text{ J g}^{-1}^\circ\text{C}^{-1}$.
15. 30 moles of nitrogen gas is kept in a canister at a temperature of 47°C . Due to a leak in the canister the pressure drops to 75% of its original value and the temperature drops to 27°C . Assuming nitrogen to be an ideal gas, the amount of nitrogen (in moles) that has leaked out is :
16. When we compare the weight of a sample to the weight measured in a magnetic field, paramagnetic samples that are attracted to the magnet will appear heavier because of the force exerted by the magnetic field. How many of the following substances will appear heavier if weight is measured in a magnetic field?
 B_2 , C_2 , Na , KO_2 , NO_2 , H_2O , NaCl , NO , ClO_2 , O_2
17. o-xylene is a hydrocarbon having structural formula as shown below.



In o-xylene X_1 is maximum number of atoms in a plane, X_2 is maximum number of sp^2 hybridised atoms and X_3 is maximum number of σ bonds. Find the value of $X_1 + X_2 + X_3$.

18. How many of the following are linear and polar?

N_2O , O_3 , SO_2 , XeF_2 , CO_2 , SF_2 , BeF_2 , SiF_2 , NO_2

SPACE FOR ROUGH WORK

PART III : MATHEMATICS**60 MARKS****SECTION 1**

This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- Straight lines $2x + y = 5$ and $x - 2y = 3$ intersect at the point A. Points B and C are chosen on these two lines such that $AB = AC$. Then the equation of a line BC passing through the point (2, 3) is:

(A) $3x - y - 3 = 0$ (B) $x + 3y - 11 = 0$ (C) $3x + y - 9 = 0$ (D) $x - 3y + 7 = 0$
- Which of the following must hold good for the expansion of the binomial $\left(x^4 + \frac{1}{x^3}\right)^{15}$?

(A) There exists a term which is independent of x .
 (B) 8th and 9th terms of the expansion have the greatest binomial coefficient.
 (C) Coefficients of x^{32} and x^{-17} are equal
 (D) If $x = \sqrt{2}$ then number of rational terms in the expansion is 5.
- A man has 3 friends. Number of ways in which he can invite one friend every day for dinner on 6 successive nights so that he invites every friend at least once is N . Which of the following hold(s) good?

(A) Number of divisors of N which are divisible by 5 is 12
 (B) Sum of all the divisors of N is 1680
 (C) Number of ways in which N can be resolved as product of two divisors which are relatively prime, is 4
 (D) Number of divisors of the form $(4n + 2), n \in N$ is 6.
- Words are formed using all the letters of the word $W = POTATO$. Let
 'a' denote the numbers of words when all the vowels are together.
 'b' denote the number of words when vowels as well as consonant are separated and
 'c' denote the number of words which begin and end in vowels.
 Which of the following is(are) correct?

(A) $a > c > b$ (B) $a + c = 4b$ (C) $a + b + c = 90$ (D) $3b = 2c$
- Let $B(1, -3)$ and $D(0, 4)$ represent two vertices of rhombus ABCD in (x, y) plane, then coordinates of vertex A if $\angle BAD = 60^\circ$ can be equal to:

(A) $\left(\frac{1-7\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}\right)$ (B) $\left(\frac{1+7\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$
 (C) $\left(\frac{1-14\sqrt{3}}{2}, \frac{1-2\sqrt{3}}{2}\right)$ (D) $\left(\frac{1+14\sqrt{3}}{2}, \frac{1+2\sqrt{3}}{2}\right)$
- If $S_n = {}^nC_0 \cdot {}^nC_1 + {}^nC_1 \cdot {}^nC_2 + {}^nC_2 \cdot {}^nC_3 + \dots + {}^nC_{n-1} \cdot {}^nC_n$ then, for very large value of n (n approaches infinity)

(A) $\frac{S_{n+1}}{S_n} = 4$ (B) $\frac{S_{n+1}}{S_n} = 2$ (C) $\frac{S_{n+3}}{S_n} = 64$ (D) $\frac{S_{n+3}}{S_n} = 8$

SPACE FOR ROUGH WORK

SECTION 2

This section has TWO (02) Paragraphs containing TWO (02) Questions on each paragraph. Each question has FOUR options. ONLY ONE of these four options is the correct answer.

Paragraph for Q-7 to Q-8

16 players $P_1, P_2, P_3, \dots, P_{16}$ take part in a tennis tournament. Lower subscript player is better than any higher subscript player. These players are to be divided into 4 groups each comprising of 4 players and the best from each group is selected for semifinals.

7. Number of ways in which 16 players can be divided into four equal groups, is:

(A) $\frac{35}{27} \prod_{r=1}^8 (2r-1)$ (B) $\frac{35}{24} \prod_{r=1}^8 (2r-1)$ (C) $\frac{35}{52} \prod_{r=1}^8 (2r-1)$ (D) $\frac{35}{6} \prod_{r=1}^8 (2r-1)$

8. Number of ways in which these 16 players can be divided into four equal groups, such that when the best player is selected from each group, P_6 is one among them, is $(k) \frac{12!}{(4!)^3}$. The value of k is:

(A) 36 (B) 24 (C) 18 (D) 20

Paragraph for Q-9 to Q-10

A straight line passing through $O(0,0)$ cuts the lines $x = \alpha, y = \beta$ and $x + y = 8$ at A, B and C respectively such that $OA \cdot OB \cdot OC = 48\sqrt{2}$ and $f(\alpha, \beta) \leq 0$

Where $f(x, y) = \left| \frac{y}{x} - \frac{3}{2} \right| + (3x - 2y)^6 + \sqrt{ex + 2y - 2e - 6}$

9. $OA + OB + OC$ equals:

(A) $2\sqrt{2}$ (B) $4\sqrt{2}$ (C) $7\sqrt{2}$ (D) $9\sqrt{2}$

10. Equation of line OA is :

(A) $x - y = 0$ (B) $x - \sqrt{2}y = 0$ (C) $\sqrt{2}x - y = 0$ (D) $2x - y = 0$

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SECTION 3

This section contains 2 Column-Match sets. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

11. Match the Column I and Column II

Column I		Column II	
(I)	The remainder when 9^{103} is divided by 25 is equal to	(P)	7
(II)	$\sum_{r=1}^n \left\{ \sum_{r_1=0}^{r-1} {}^nC_r {}^rC_{r_1} 2^{r_1} \right\}$ is equal to $x^n - y^n$ then, $x + y$ is equal to	(Q)	2
(III)	If the sum of the coefficients in the expansion of $(1+2x)^n$ is 6561, the greatest term in the expansion for $x = \frac{1}{2}$ is k^{th} term then, k is equal to	(R)	4
(IV)	If $P = (6\sqrt{6} + 14)^{2n+1}$ and $f = P - [P]$, where $[]$ denotes the greatest integer function, then Pf equals k^{2n+1} , then $k/10$ equals	(S)	5

Code :

- (A) I-S ; II-P ; III-Q ; IV-R (B) I-R ; II-S ; III-S ; IV-Q
(C) I-R ; II-P ; III-S ; IV-Q (D) I-S ; II-Q ; III-R ; IV-P

12. Consider the system of 3 linear equations

$$ax + by + c = 0$$

$$bx + cy + a = 0$$

$$cx + ay + b = 0$$

where $a, b, c \in R$.

Column 1		Column 2	
(A)	If $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$, then the system represents	(P)	Entire xy plane
(B)	If $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$, then the system represents	(Q)	Concurrent lines
(C)	If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$, then the system represents	(R)	Coincident lines
(D)	If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$, then the system represents	(S)	Neither coincident nor concurrent lines

(A) [A-R] [B-Q] [C-P] [D-R]

(B) [A-Q] [B-Q] [C-P] [D-Q]

(C) [A-Q] [B-P] [C-S] [D-R]

(D) [A-R] [B-P] [C-S] [D-Q]

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SECTION 4

This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.*

(Example: 6, 81, 1.50, 3.25, 0.08)

13. If the binomial coefficients of 2^{nd} , 3^{rd} and 4^{th} terms in the expansion of $\left[\sqrt{2^{\log_{10}(10-3^x)}} + \sqrt[5]{2^{(x-2)\log_{10}3}} \right]^m$ are in AP and the 6^{th} term is 21, then the larger value of x is _____.
14. Triangle ABC with $AB = 13$, $BC = 5$ and $AC = 12$ slides on the coordinate axes with A and B on the positive x-axis and positive y-axis respectively, while C always lies in the first quadrant. If vertex C always lies on the line $12x - ky = 0$, then the value of k is _____.
15. $\sum_{r=1}^{20} \left(r^2 \left({}^{20}C_r - {}^{20}C_{r-1} \right) + (2r+1) {}^{20}C_r \right) = A$, sum of digits of A is _____.
16. 10 identical balls are distributed in 5 different boxes kept in a row and labeled A, B, C, D and E. Find the number of ways in which the ball can be distributed in the boxes if no two adjacent boxes remain empty.
17. In how many ways it is possible to select six letters, including at least one vowel from the letters of the word "F L A B E L L I F O R M".
18. If the equations of the diagonals of the parallelogram formed by the lines, $2x - y + 7 = 0$; $2x - y - 5 = 0$; $3x + 2y - 5 = 0$ and $3x + 2y + 4 = 0$ are $ax + by - 5 = 0$ and $px + qy + 1 = 0$, where a, b, p, q are natural number. Find the value of $(a + b + p + q)$.

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